

Problem

A series RLC network has $R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, and $C = 1 \text{ }\mu\text{F}$. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

Step-by-step solution

Step 1 of 6

Calculate the impedance of the series RLC circuit.

$$\begin{aligned} Z &= Z_R + Z_L + Z_C \\ &= R + j\omega L + \frac{1}{j\omega C} \\ &= R + j\left(\omega L - \frac{1}{\omega C}\right) \end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned} \omega_o &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(40 \times 10^{-3})(1 \times 10^{-6})}} \\ &= 5 \text{ krad/s} \end{aligned}$$

Step 2 of 6

Calculate the impedance of the circuit at resonance, $\omega = \omega_o$.

$$\begin{aligned} Z(\omega_o) &= R + j\left(\omega_o L - \frac{1}{\omega_o C}\right) \\ &= 2000 + j\left((5 \times 10^3)(40 \times 10^{-3}) - \frac{1}{(5 \times 10^3)(1 \times 10^{-6})}\right) \\ &= 2000 + j(200 - 200) \\ &= 2 \text{ k}\Omega \end{aligned}$$

Thus, the impedance at resonance is $\boxed{2 \text{ k}\Omega}$.

Step 3 of 6

Calculate the impedance of the circuit at one-fourth of the resonance, $\omega = \frac{\omega_o}{4}$.

$$\begin{aligned} Z\left(\frac{\omega_o}{4}\right) &= R + j\left(\frac{\omega_o}{4} L - \frac{4}{\omega_o C}\right) \\ &= 2000 + j\left(\frac{5 \times 10^3}{4}(40 \times 10^{-3}) - \frac{4}{(5 \times 10^3)(1 \times 10^{-6})}\right) \\ &= 2000 + j\left(50 - \frac{4000}{5}\right) \\ &= 2 - j0.75 \text{ k}\Omega \end{aligned}$$

Thus, the impedance at one-fourth of the resonance is $\boxed{2 - j0.75 \text{ k}\Omega}$.

Step 4 of 6

Calculate the impedance of the circuit at one-half of the resonance, $\omega = \frac{\omega_0}{2}$.

$$\begin{aligned} Z\left(\frac{\omega_0}{2}\right) &= R + j\left(\frac{\omega_0}{2}L - \frac{2}{\omega_0 C}\right) \\ &= 2000 + j\left(\frac{5 \times 10^3}{2}(40 \times 10^{-3}) - \frac{2}{(5 \times 10^3)(1 \times 10^{-6})}\right) \\ &= 2000 + j\left(100 - \frac{2000}{5}\right) \\ &= 2 - j0.3 \text{ k}\Omega \end{aligned}$$

Thus, the impedance at one-half of the resonance is $\boxed{2 - j0.3 \text{ k}\Omega}$.

Step 5 of 6

Calculate the impedance of the circuit at twice of the resonance, $\omega = 2\omega_0$.

$$\begin{aligned} Z(2\omega_0) &= R + j\left(2\omega_0 L - \frac{1}{2\omega_0 C}\right) \\ &= 2000 + j\left(2(5 \times 10^3)(40 \times 10^{-3}) - \frac{1}{2(5 \times 10^3)(1 \times 10^{-6})}\right) \\ &= 2000 + j\left(400 - \frac{1000}{10}\right) \\ &= 2 + j0.3 \text{ k}\Omega \end{aligned}$$

Thus, the impedance at twice the resonance is $\boxed{2 + j0.3 \text{ k}\Omega}$.

Step 6 of 6

Calculate the impedance of the circuit at four times the resonance, $\omega = 4\omega_0$.

$$\begin{aligned} Z(4\omega_0) &= R + j\left(4\omega_0 L - \frac{1}{4\omega_0 C}\right) \\ &= 2000 + j\left(4(5 \times 10^3)(40 \times 10^{-3}) - \frac{1}{4(5 \times 10^3)(1 \times 10^{-6})}\right) \\ &= 2000 + j\left(800 - \frac{1000}{20}\right) \\ &= 2 + j0.75 \text{ k}\Omega \end{aligned}$$

Thus, the impedance at four times the resonance is $\boxed{2 + j0.75 \text{ k}\Omega}$.